

ApexCare Health Solutions

Healthcare Insurance Prior Authorization Assistant

TESTING POLICY MANUAL & GENAI EVALUATION DATASET

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Purpose: Fictional policy dataset for testing Retrieval-Augmented Generation (RAG), document understanding, policy retrieval, and structured prior authorization summaries.

IMPORTANT: This is a synthetic testing document. It is not a real insurance policy and must not be used for actual medical, insurance, or treatment decisions.

1. Problem Scope

Healthcare prior authorization requires a provider, hospital, or member to obtain approval before selected services are covered. The GenAI assistant should analyze the authorization request against the policy and generate a structured summary without inventing missing information.

Expected Assistant Output

Field	Expected Content
Authorization Status	Recommended / Not Recommended / Pending Additional Information
Requested Service	Treatment, procedure, imaging, surgery, or medication
Coverage Determination	Covered / Non-Covered / Conditional
Criteria Met	Requirements supported by submitted information
Criteria Not Met	Policy requirements that failed
Missing Information	Documentation required before determination
Exclusions	Applicable exclusions or contraindications
Waiting Periods	Applicable time-based restrictions
Policy Reference	Relevant section and rule
Summary	Short explanation grounded in policy text

2. General Prior Authorization Governance

Prior authorization is mandatory for services explicitly listed in this document. Requests must contain sufficient clinical documentation to evaluate medical necessity. An authorization request is not automatically approved merely because the requested service appears in the covered-services list.

General Documentation Requirements:

- Treating provider chart notes dated within the previous 60 days.
- Relevant diagnostic reports and dates of service.
- Diagnosis and clinical indication.

- Documentation of conservative treatment where required.
- Medication history and prior treatment failures where applicable.
- Objective clinical findings when continuation or repeat services are requested.

3. Covered Treatments and Procedures

3.1 Advanced Diagnostic Imaging - Lumbar Spine MRI

Applicable CPT Codes: 72148, 72149, 72158

Lumbar Spine MRI is conditionally covered when **ANY** of the following criteria is met:

1. **Red Flag Pathology:** Suspected cauda equina syndrome, progressive major motor neurologic deficit, active malignancy with bone metastasis risk, or documented spinal infection.
2. **Conservative Therapy Failure:** Persistent lower back pain with or without radicular symptoms lasting at least 6 consecutive weeks AND documentation of at least 6 continuous weeks of supervised physical therapy or structured NSAID/analgesic treatment without adequate relief.
3. **Pre-Surgical Planning:** Candidate is scheduled for immediate spinal surgical decompression within 30 days.

3.2 Specialty Biologics - Adalimumab

Applicable Drug Codes: J0135, Q5115

Applicable Diagnosis Families: M05, M06

Adalimumab is conditionally covered for moderate to severe rheumatoid arthritis when step therapy and baseline testing requirements are met.

Step Therapy: Documented failure of Methotrexate at a minimum dose of 15 mg weekly for at least 12 consecutive weeks, OR documented contraindication/intolerance to Methotrexate PLUS a 12-week trial of Leflunomide or Sulfasalazine.

3.3 Total Knee Arthroplasty

Applicable CPT Code: 27447

Applicable ICD-10 Range: M17.0 - M17.9

Total knee arthroplasty is conditionally covered when **ALL** of the following criteria are met:

1. Advanced osteoarthritis or joint destruction confirmed by weight-bearing radiographs showing Kellgren-Lawrence Grade 3 or 4 within the preceding 12 months.
2. Persistent functional impairment severely limiting daily activities.
3. Failure of at least 3 months of non-operative conservative management.
4. Compliance with at least 2 approved non-surgical modalities.

4. Conservative Treatment and Step Therapy Requirements

4.1 Approved Non-Surgical Modalities for Knee Arthroplasty

Modality	Minimum Requirement
Physical Therapy	Formal PT or supervised home exercise; minimum 8 sessions
Injection Therapy	Intra-articular corticosteroid or hyaluronic acid injection
Medication Therapy	Systemic NSAIDs or scheduled analgesics for at least 6 weeks
Assistive Device	Documented cane, walker, or similar ambulation device use

4.2 Rheumatoid Arthritis Step Therapy

Scenario	Required Evidence
Methotrexate failure	15 mg weekly minimum for 12 consecutive weeks with inadequate response
Methotrexate intolerance	Documented adverse reaction or contraindication
Alternative csDMARD	Leflunomide 20 mg/day or Sulfasalazine 2 g/day for at least 12 weeks
Biologic continuation	Objective clinical improvement required for renewal

4.3 Required Baseline Testing for Adalimumab

- Negative Tuberculin PPD or QuantiFERON-TB test documented within the preceding 12 months.
- Hepatitis B screening including HBsAg and anti-HBc before therapy initiation.
- Current diagnosis and disease severity documentation.
- Prior medication and treatment history.

5. Non-Covered Services

The following are non-covered under this synthetic testing policy unless another explicit exception is documented:

- Cosmetic procedures performed solely for appearance enhancement.
- Experimental or investigational procedures without policy exception.
- Routine imaging without documented medical necessity.
- Duplicate diagnostic testing without new symptoms or clinical findings.
- Services requested with insufficient supporting documentation.
- Procedures performed before mandatory authorization when prior authorization was required.

6. Explicit Exclusions and Restrictions

6.1 Lumbar Spine MRI Exclusions

- Uncomplicated acute lower back pain lasting less than 6 weeks without neurologic compromise.
- Routine surveillance imaging in asymptomatic post-operative individuals.
- Repeat MRI within 6 months unless new focal neurological deficits or material clinical changes are documented.

6.2 Adalimumab Exclusions

- Missing tuberculosis screening.
- Missing Hepatitis B baseline testing.
- Failure to complete required step therapy without documented contraindication.
- Continuation request without documented objective clinical improvement.
- Request submitted without diagnosis supporting moderate to severe rheumatoid arthritis.

6.3 Total Knee Arthroplasty Exclusions and Contraindications

- Active local joint infection or systemic bacteremia.
- Severe peripheral arterial occlusive disease in the operative limb.
- BMI greater than or equal to 45 without documented bariatric or supervised medical weight-reduction clearance.
- Absence of qualifying radiographic evidence.
- Less than 3 months of documented conservative management.

7. Waiting Periods and Time-Based Rules

Service / Rule	Time Requirement
Lumbar pain conservative therapy	At least 6 consecutive weeks
Lumbar MRI repeat restriction	No repeat within 6 months unless new deficits
Methotrexate trial	At least 12 consecutive weeks
Alternative csDMARD trial	At least 12 consecutive weeks
Knee conservative management	At least 3 months
Knee radiograph validity	Within previous 12 months
Provider chart notes	Within previous 60 days
Adalimumab initial approval	6 months
TB screening validity	Within previous 12 months

8. GenAI Assistant Decision Rules

The assistant should retrieve relevant policy text and produce a determination based only on documented information and policy criteria.

Recommended Status Definitions:

Status	Meaning
Recommended	Available documentation satisfies applicable policy criteria and no exclusion is identified
Not Recommended	Available documentation clearly fails one or more mandatory policy requirements or an exclusion applies
Pending Additional Information	The request may qualify, but required information is missing or ambiguous

Mandatory Assistant Behavior

1. Do not invent clinical facts.
2. Do not assume undocumented therapy was completed.
3. Identify the exact missing evidence.
4. Separate coverage eligibility from clinical treatment advice.
5. Cite the relevant policy section.
6. When multiple rules apply, evaluate all applicable rules.
7. If an exclusion applies, clearly state it.
8. Use 'Pending Additional Information' rather than guessing when evidence is incomplete.

Recommended Structured Output Template

Authorization Status: [Recommended / Not Recommended / Pending Additional Information]

Requested Service: ...

Coverage Classification: Covered / Conditional / Non-Covered

Criteria Met: ...

Criteria Not Met: ...

Missing Information: ...

Applicable Exclusions: ...

Waiting Periods / Time Rules: ...

Policy Reference: ...

Summary: ...

9. Test Cases for GenAI Evaluation

These synthetic cases are designed to test retrieval accuracy, policy reasoning, missing-information detection, exclusions, and time-based criteria.

Test Case 1 - Acute Back Pain MRI

Request: Patient has lower back pain for 3 weeks. No neurological deficits, no malignancy, no infection, and no physical therapy completed. Lumbar MRI requested.

Expected Status: Not Recommended

Expected Reason: Pain duration is less than 6 weeks, no red-flag pathology is documented, and conservative therapy requirements are not met.

Policy Reference: Section 3.1 and Section 6.1

Test Case 2 - MRI After Conservative Therapy

Request: Patient has radicular lower back pain for 8 weeks and completed 7 weeks of supervised physical therapy without adequate relief.

Expected Status: Recommended

Expected Reason: Duration and conservative therapy requirements are met.

Policy Reference: Section 3.1

Test Case 3 - Emergency Red Flag MRI

Request: Patient presents with suspected cauda equina syndrome and progressive lower extremity weakness. Symptoms began 2 days ago.

Expected Status: Recommended

Expected Reason: Red-flag pathology independently qualifies for MRI; the 6-week conservative therapy requirement does not apply.

Policy Reference: Section 3.1

Test Case 4 - Repeat MRI

Request: Patient requests repeat lumbar MRI 3 months after previous MRI. No new neurological deficits or material clinical changes are documented.

Expected Status: Not Recommended

Expected Reason: Repeat MRI is restricted within 6 months without new focal neurological findings.

Policy Reference: Section 6.1

Test Case 5 - Adalimumab With Full Step Therapy

Request: Patient has moderate rheumatoid arthritis. Methotrexate 15 mg weekly was used for 14 weeks without adequate response. TB and Hepatitis B tests are documented.

Expected Status: Recommended

Expected Reason: Required Methotrexate trial and baseline testing are documented.

Policy Reference: Section 3.2 and Section 4.2

Test Case 6 - Biologic Missing TB Test

Request: Patient completed Methotrexate for 16 weeks with treatment failure. Hepatitis B testing is available, but no TB screening result is submitted.

Expected Status: Pending Additional Information

Expected Reason: TB screening is mandatory before therapy initiation.

Policy Reference: Section 4.3

Test Case 7 - Biologic Insufficient Step Therapy

Request: Patient used Methotrexate for only 4 weeks and stopped because they felt it was not helping. No documented adverse reaction or contraindication.

Expected Status: Not Recommended

Expected Reason: Required 12-week trial was not completed and no documented exception exists.

Policy Reference: Section 4.2

Test Case 8 - Knee Replacement Fully Qualified

Request: Patient has Kellgren-Lawrence Grade 4 osteoarthritis confirmed 4 months ago, severe mobility limitations, 5 months of conservative treatment, 12 PT sessions, and corticosteroid injection.

Expected Status: Recommended

Expected Reason: All mandatory criteria are met, including radiographic evidence, functional impairment, duration, and two conservative modalities.

Policy Reference: Section 3.3 and Section 4.1

Test Case 9 - Knee Replacement Missing Imaging

Request: Patient has severe knee pain and completed 6 months of conservative treatment with PT and NSAIDs. No weight-bearing radiograph report is provided.

Expected Status: Pending Additional Information

Expected Reason: Qualifying radiographic evidence is mandatory but missing.

Policy Reference: Section 3.3

Test Case 10 - Knee Replacement Infection

Request: Patient meets radiographic and conservative treatment requirements but currently has an active local knee joint infection.

Expected Status: Not Recommended

Expected Reason: Active local joint infection is an exclusion/contraindication.

Policy Reference: Section 6.3

Test Case 11 - Knee BMI Restriction

Request: Patient meets all TKA criteria and has BMI 47. No bariatric clearance or supervised medical weight-reduction clearance is documented.

Expected Status: Not Recommended

Expected Reason: BMI restriction applies and required clearance is absent.

Policy Reference: Section 6.3

Test Case 12 - Old Provider Notes

Request: Patient otherwise qualifies for MRI, but the only treating provider chart notes are dated 90 days ago.

Expected Status: Pending Additional Information

Expected Reason: Current policy requires chart notes within the preceding 60 days.

Policy Reference: Section 2

Test Case 13 - Cosmetic Procedure

Request: Member requests authorization for a cosmetic procedure solely to improve physical appearance.

Expected Status: Not Recommended

Expected Reason: Cosmetic procedures performed solely for appearance enhancement are non-covered.

Policy Reference: Section 5

Test Case 14 - Ambiguous RA Severity

Request: Provider requests Adalimumab. Prior Methotrexate failure and baseline tests are documented, but rheumatoid arthritis severity is not stated.

Expected Status: Pending Additional Information

Expected Reason: Diagnosis severity supporting moderate to severe RA must be documented.

Policy Reference: Section 3.2 and Section 6.2

Test Case 15 - Conflicting Documentation

Request: Request states patient completed 12 weeks of Methotrexate, but submitted medication records show only 6 weeks.

Expected Status: Pending Additional Information

Expected Reason: Conflicting evidence must be clarified; the assistant should not choose one unsupported version.

Policy Reference: Section 5 and Section 8

10. Evaluation Checklist for Your GenAI Assistant

Capability	Pass Condition
Document Retrieval	Retrieves the correct policy section for each request
Coverage Detection	Distinguishes covered, conditional, and non-covered services
Criteria Evaluation	Correctly checks ALL versus ANY requirements
Exclusion Detection	Identifies applicable exclusions and contraindications
Waiting Period Reasoning	Correctly evaluates durations and date windows
Missing Information	Lists specific missing documents instead of guessing
Conflict Detection	Flags contradictory evidence
Hallucination Control	Does not invent clinical facts or policy rules
Structured Output	Returns all requested fields consistently
Policy Grounding	References the relevant policy section

11. Suggested Evaluation Metrics

For automated testing, compare the assistant's response against expected outcomes using the following metrics:

- **Authorization Accuracy:** Correct status prediction.
- **Retrieval Accuracy:** Correct policy section retrieved.
- **Criteria Precision:** Correct identification of met and unmet criteria.
- **Missing Information Accuracy:** Correctly identifies required documents.
- **Exclusion Recall:** Detects all applicable exclusions.
- **Hallucination Rate:** Measures unsupported statements.
- **Output Completeness:** Measures whether all structured fields are present.

12. Final Testing Instruction

For each authorization request, provide the policy manual to the GenAI system through direct context or a RAG pipeline. Ask the system to generate only the structured prior authorization summary. Compare the response with the expected test-case result. A strong system should explain the determination using policy evidence and explicitly state when additional information is required.

END OF SYNTHETIC TESTING POLICY MANUAL